Static Ledger: Boolean Inverse Algebra over $GF(4)$

Exact Complement, Gate Reduction, and Inverse-Metric Closure

Foliation Engine — Theoretical Formalization (W10)

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Abstract

We record a purely algebraic settlement of finite-field Boolean inversion on $GF(4) \cong GF(2^2)$. Sequence symbols are identified with field elements; the exact complement equation $\mathbf{S}_{\text{path}} \oplus \mathbf{S}_{\text{thera}} = \mathbf{1}$ is solved by a depth-$O(1)$ Boolean map on the natural $GF(2)^2$ bit lift; macrocyclic closure is stated as an exact residual $\Delta E = 0$ under an inverse Hermitian metric constraint. No wet-lab protocol, no statistical estimator, and no affinity model appear in the ledger.

1 Galois Field and Symbol Isomorphism

Definition 1 (Binary extension field)**.** *Let $\mathbb{F}_2 = \{0, 1\}$ and let*

$$GF(4) = GF(2^2) = \mathbb{F}_2[x]/(x^2 + x + 1).$$

Write α for the residue class of x. Then $\{0, 1, \alpha, \alpha^2\}$ is the additive group of $GF(4)$, with $\alpha^2 + \alpha + 1 = 0$ and $\alpha^3 = 1$.

Definition 2 (Canonical symbol isomorphism)**.** *Define $\iota : \{A, U, C, G\} \to GF(4)$ by*

$$\iota(A) = 0, \qquad \iota(U) = 1, \qquad \iota(C) = \alpha, \qquad \iota(G) = \alpha^2. \tag{1}$$

Equivalently, under the ordered bit lift $\varphi : GF(4) \to GF(2)^2$ given by $\varphi(b_0 + b_1\alpha) = (b_0, b_1)$,

$$\begin{aligned} \iota(A) &\mapsto (0, 0), \quad \iota(U) \mapsto (1, 0), \\ \iota(C) &\mapsto (0, 1), \quad \iota(G) \mapsto (1, 1). \end{aligned} \tag{2}$$

Remark 1. *The map ι is a set bijection, not a ring homomorphism from a nucleotide monoid. All subsequent identities are identities in $GF(4)^N$ or in the bit lift $GF(2)^{2N}$.*

Definition 3 (Path vector)**.** *For length N, a path state is a vector $\mathbf{S}_{\text{path}} = (s_i)_{i=0}^{N-1} \in GF(4)^N$. Its bit lift is $\mathbf{B} = (b_{i,0}, b_{i,1})_{i=0}^{N-1} \in GF(2)^{2N}$ with $\varphi(s_i) = (b_{i,0}, b_{i,1})$.*

2 Exact-Form Boolean Complement

Definition 4 (Field XOR and the ledger one-vector)**.** *Write $\oplus$ for coordinatewise addition in $GF(4)$. Let the ledger one-vector be the constant vector of the primitive element α:*

$$\mathbf{1} := (\alpha, \alpha, \dots, \alpha)^\top \in GF(4)^N. \tag{3}$$

(The multiplicative unit of $GF(4)$ is written $1_\times$ when needed; it is not the target of (4).)

Theorem 1 (Unique Boolean inverse). *For every* $\mathbf{S}_{\text{path}} \in GF(4)^N$ *there exists a unique* $\mathbf{S}_{\text{thera}} \in GF(4)^N$ *such that*

$$\mathbf{S}_{\text{path}} \oplus \mathbf{S}_{\text{thera}} = \mathbf{1}. \tag{4}$$

Explicitly,

$$\mathbf{S}_{\text{thera}} = \mathbf{1} \oplus \mathbf{S}_{\text{path}}.$$

Proof. $(GF(4), +)$ is an elementary abelian 2-group, so each coordinate equation $s_i \oplus t_i = \alpha$ has the unique solution $t_i = \alpha \oplus s_i$. Vector uniqueness follows coordinatewise. □

Proposition 1 (Bit-level gate reduction). *Under the bit lift* $\varphi(b_0 + b_1\alpha) = (b_0, b_1)$*, equation* (4) *is equivalent to the Boolean system*

$$t_{i,0} = b_{i,0}, \qquad t_{i,1} = \neg b_{i,1} \qquad (i = 0, \dots, N-1), \tag{5}$$

where $\neg$ *denotes complement in* $GF(2)$ *and* $\varphi(t_i) = (t_{i,0}, t_{i,1})$.

Proof. Write $s_i = b_{i,0} + b_{i,1}\alpha$. Then

$$\alpha \oplus s_i = (0 \oplus b_{i,0}) + (1 \oplus b_{i,1})\alpha = b_{i,0} + (\neg b_{i,1})\,\alpha,$$

hence $\varphi(\alpha \oplus s_i) = (b_{i,0}, \neg b_{i,1})$, which is exactly (5). The four-row truth table is:

s	(b_0, b_1)	$\alpha \oplus s$	(t_0, t_1)
0	$(0,0)$	α	$(0,1)$
$1_\times$	$(1,0)$	α^2	$(1,1)$
α	$(0,1)$	0	$(0,0)$
α^2	$(1,1)$	$1_\times$	$(1,0)$.

□

Corollary 1 (Combinational complexity). *The map* $\mathbf{S}_{\text{path}} \mapsto \mathbf{S}_{\text{thera}}$ *is a coordinatewise Boolean function of circuit depth* $O(1)$ *and size* $O(N)$*. No iteration or search is required.*

3 Inverse-Metric Constraint and Exact Closure

Definition 5 (Clearance slack and inverse metric). *Let* $q \in \mathbb{R}^{3m}$ *collect Cartesian coordinates of* m *atoms in a closed macromolecular graph. For each steric pair* e *with clearance slack* $\sigma_e(q) > 0$*, define the local metric weight*

$$g_e(q) = \frac{\mu L^2}{\sigma_e(q)^2}, \qquad \mu > 0,\ L > 0, \tag{6}$$

and the inverse weight $g_e^{-1}(q) = 1/g_e(q)$*. Collecting weights into a diagonal metric* $G(q)$ *yields the inverse*

$$G^{-1}(q) = \operatorname{diag}\big(g_e^{-1}(q)\big). \tag{7}$$

This G^{-1} *is the static ledger's inverse Hermitian (Kähler-type) brake:* $g_e^{-1} \to 0$ *as* $\sigma_e \to 0$.

Definition 6 (Closure residual). *Let* $\mathcal{T}$ *embed a* $GF(4)$ *state into a torsional chart of the closed graph, and let* $E(q)$ *denote the exact-form cycle residual (discrete circulation of edge lengths about each independent cycle). Exact closure is the null statement*

$$\Delta E := E(q) = 0. \tag{8}$$

Theorem 2 (Coupled algebraic–geometric settlement)**.** *Suppose* $\mathbf{S}_{\text{thera}}$ *satisfies* (4) *and* $q^\star$ *lies in the feasible open set* $\{\sigma_e > 0\}$*. If the inverse metric annihilates the coupled tensor chart,*

$$\mathcal{H}^{-1}(\mathcal{T}(\mathbf{S}_{\text{path}}) \otimes \mathcal{T}(\mathbf{S}_{\text{thera}})) = \mathbf{0}, \tag{9}$$

and the discrete cycle operator enforces (8)*, then the pair* $(\mathbf{S}_{\text{thera}}, q^\star)$ *is an exact algebraic–geometric settlement: Boolean complement holds identically, and macrocyclic closure holds with residual zero.*

Proof. Equation (4) is already settled by Theorem 1. Feasibility $\sigma_e(q^\star) > 0$ keeps G^{-1} finite. Condition (9) forces the coupled chart into the kernel of the inverse metric brake. Independently, $\Delta E = 0$ is the definition of exact-form null for cyclic edge residuals. The conjunction is therefore a static settlement certificate in the product space $GF(4)^N \times \mathbb{R}^{3m}$. □

Proposition 2 (No affinity content)**.** *The identity* $\Delta E = 0$ *is a geometric closure constraint. It does not imply a free-energy ranking, a scoring-function minimum, or a statistical association constant.*

4 Static Settlement Ledger

Object	Symbol	Status
Field	$GF(4) \cong \mathbb{F}_2[x]/(x^2 + x + 1)$	fixed
Isomorphism	ι	fixed
Complement	$\mathbf{S}_{\text{path}} \oplus \mathbf{S}_{\text{thera}} = \mathbf{1}$	unique
Gate reduction	$t_{i,0} = b_{i,0},\ t_{i,1} = \neg b_{i,1}$	$O(1)$ depth
Inverse metric	$G^{-1} \to 0$ as slack $\to 0$	brake
Closure	$\Delta E = 0$	exact null

Disclaimer

This document is a mathematical ledger only (`clinical_grade=false`). It encodes finite-field identities and geometric closure constraints. It does not claim therapeutic effect, manufacturing readiness, or empirical binding.